

Name: \_\_\_\_\_

Circle your teacher: FEUTRILL FRIDAY HILL LOH MARTIN

**CALCULATOR ASSUMED SECTION** (Total marks: 38 marks)

One A4 page, single sided notes allowed.

Show full working to obtain full marks for each of the questions in this section.

All answers should be given in two decimal places unless otherwise stated.

Time: 35 minutes

**QUESTION 1 [4 marks – 2, 2]**

Soccer balls come in different sizes for different age group. A Size 3 soccer ball has a circumference of 58 cm while a Size 5 soccer ball has a circumference of 70 cm.

a) If the volume of a Size 5 soccer ball is 5 580 cm<sup>3</sup>, calculate the volume of the smaller soccer ball to the nearest cm<sup>3</sup>.

$$C_3 : C_5 = 58 : 70$$

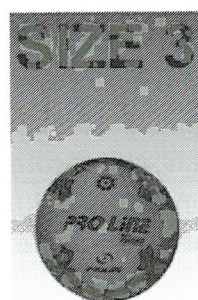
$$V_3 : V_5 = (58)^3 : (70)^3$$

$$V_3 : 5580 = (58)^3 : (70)^3$$

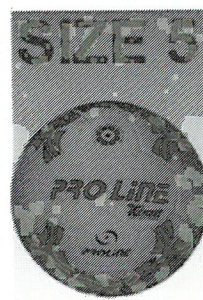
$$V_3 = \frac{(58)^3}{(70)^3} 5580$$

$$= 3174.13$$

$$= 3174 \text{ cm}^3$$



Under 8 years of age



12 years of age and beyond  
Official Size and Weight

correct ratio —1  
correct answer —1

b) If the surface area of the Size 3 soccer ball is 1 020 cm<sup>2</sup>, calculate the surface area of the bigger soccer ball to the nearest cm<sup>2</sup>.

$$S_3 : S_5 = (58)^2 : (70)^2$$

$$S_3 = 1020$$

$$S_5 = 1020 \frac{(70)^2}{(58)^2}$$

$$= 1485.73$$

$$= 1486 \text{ cm}^2$$

correct ratio —1  
correct answer —1

—1 overall for incorrect rounding  
—1 overall for incorrect unit or no unit

7 2 4  
QUESTION 7 [6 marks - 3, 1, 2]

Eric made a trip to visit a client. The traffic he encountered kept his average speed to 40 km/h. On the return trip, he took a route 6 km longer, but he averaged 50 km/h. If he took the same time each way, how long was the total journey? (Hint: Distance = speed x time).

a) Write an expression to represent Eric's initial trip to visit the client, clearly defining the variables you use.

let  $x$  = distance for the first trip ✓  
 $t$  = time travelled by Eric ✓

$$x = 40t \quad \checkmark$$

b) Using the variables you have defined in part a, write an expression to represent his return trip.

$$x + 6 = 50t \quad \checkmark$$

c) Hence calculate the total distance travelled by Eric.

from (a)  $t = \frac{x}{40}$   
(b)  $t = \frac{x+6}{50}$  } rearrange ✓

equating (a) & (b)

$$\frac{x}{40} = \frac{x+6}{50} \quad \checkmark \quad \text{equation}$$

$$50x = 40x + 240$$

$$10x = 240$$

$$x = 24 \quad \checkmark \quad \text{solve}$$

The total distance Eric travelled =  $24 + 24 + 6 = 54$  km ✓

correct answer